

Couple Cluster

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Introduction

Truncating The Couple cluster ansatz and its working equations

Problem We Will be addressing

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Problem we will be addressing We are interested in Solving the molecular Schrodinger equations, N electrons.

$$\psi = \psi(r_1, \dots, r_N)$$

wave function is the function of the coordinates of N-electrons.

$$H|\psi\rangle = E|\psi\rangle$$

This is a very difficult equation to solve we need to invent some kind of approximation to it.

Hartree Fork Theory

Representing the wave function by a single slated determinant.

$$|\psi\rangle \approx |\varphi_0\rangle$$

Configuration Iteration. We expressed the wave function as all possible configuration iterations.

Configuration Interaction

$$|\psi\rangle = |\varphi_0\rangle + \sum_{ia} c_i^a |\varphi_i^{ab}\rangle + \frac{1}{4} \sum_{ia} c_{ij}^{ab} |\varphi_{ij}^{ab}\rangle + \dots$$

Note: $|\varphi_0\rangle$ means Hartree Fork Slater Determinant (Ground State wave function).

$|\varphi_i^a\rangle$ = is a slater determinant where we have removed an electron from orbital i and put it in orbital a (single excitation).

$\sum_{ia} c_i^a |\varphi_i^a\rangle$ all possible single substituted configurations.

Couple Cluster Method

$$|\psi\rangle = \mathbf{e}^T |\varphi_0\rangle$$

Why Study Couple-Cluster

- * The most successful approach to accurately Solve many-electron (molecular) systems.
- * It's quite rich and comes in many different variants.
 - Tailored CC ("multi-reference method")
 - Extended CC (more elaborate, QCC).
 - Unitary CC (Hermitian, Quantum Computing). etc

Excitations

How Do we mathematically formulate couple cluster

* One-particle basis, Orbitals: $\chi_k, k = 1, \dots, K > N$

$K > N \Rightarrow$ This Means We have more orbitals than electron we want to excite into.

* Configuration interaction (CI) method: We generate determinant basis by just populating k -orbitals with N -electrons.

HF Slater determinant

$\Phi_0 = \det(\chi_j)_{j=1}^N$ (Reference Slater determinant).

We label all these determinant with ϕ_μ

$$\Phi_\mu = x_\mu \Phi_0 = \det x_{\mu_j} \in \text{Span} \{ \Phi_\mu \}$$
$$\mu_j \in \{1, 2, \dots, K\}.$$

Where μ is a multiple index that just tells you which of the orbitals was used in constructing this determinant.

$$\psi = \sum_{\mu \geq 0} C_{\mu} |\Phi_{\mu}\rangle = \sum_{\mu > 0} C_{\mu} X_{\mu} |\phi_0\rangle + C_0 |\phi_0\rangle$$

$$\psi = \left(I + \sum_{\mu > 0} C_{\mu} X_{\mu} \right) |\Phi_0\rangle$$

$$\psi = (1 + B) |\Phi_0\rangle ; \text{ where } B = \sum_{\mu \geq 0} C_{\mu} X_{\mu}$$

which is called the cluster operator.

And the energy equation becomes;

$$H(I + B) |\Phi_0\rangle = \varepsilon(I + B) |\Phi_0\rangle$$

parametrization

- CI: $\psi = \left(I + \sum_{\mu>0} C_{\mu} X_{\mu} \right) |\Phi_0\rangle$
- CC: $\psi = \underbrace{(I + B)\Phi_0}_{CI} = \underbrace{e^T \Phi_0}_{CC} = \left(1 + T + \frac{1}{2} T^2 + \dots + \frac{T^N}{N!} \right) \Phi_0$

We partition C and T according to the number of electrons they excited.

$$\begin{cases} C = \sum_{k=1}^N C_k = C_1 + \dots + C_N \\ T = \sum_{k=1}^N T_k = T_1 + \dots + T_N \end{cases}$$
$$\begin{aligned} C_1 &= T_1 \\ C_2 &= T_2 + \frac{1}{2} (T_1)^2 \\ C_3 &= T_3 + T_2 T_1 + \frac{1}{6} (T_1)^3 \\ C_4 &= \dots \end{aligned}$$

The cluster operator expansion is given by

$$T = T_1 + T_2 + T_3 + \cdots + T_N$$

$$\begin{aligned} e^T &= 1 + T + \frac{1}{2}T^2 + \frac{1}{6}T^3 + \cdots = \sum_{k=1}^N \frac{T^k}{k!} \\ &= 1 + (T_1 + T_2 + T_3 + \cdots) + \frac{1}{2}(T_1 + T_2 + \cdots)^2 + \frac{1}{6}(T_1 + \cdots)^3 + \cdots \end{aligned}$$

Here, T_1 is called a single excitation operator and T_2 is called a double excitation operator.

Rearranging the operators according to the number k of excitations

$$= 1 + T_1 + \left\{ T_2 + \frac{1}{2} T^2 \right\} + \left\{ T_1 + \frac{1}{6} T_1^3 + T_2 T_1 \right\} + \dots$$

The Schrödinger equation is given by $H|\psi\rangle = E|\psi\rangle$.

Substituting the ansatz $He^T|\Phi_0\rangle = Ee^T|\Phi_0\rangle$ yields:

$$e^{-T}He^T|\Phi_0\rangle = E|\Phi_0\rangle$$

Projecting on the reference and on the orthogonal complement

$$\langle \phi_\mu | e^{-T}He^T | \Phi_0 \rangle = E$$

.

$$\langle \phi_\mu | e^{-T}He^T | \Phi_0 \rangle = 0$$

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